

Unified Constraints on the Time-Emergent Expansion Model (TEEM):

A Joint Analysis of Type Ia Supernovae, BAO, and the Cosmic Tension Field

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Abstract

We present a unified analysis of the Time-Emergent Expansion Model (TEEM) using Type Ia supernovae (SN Ia), baryon acoustic oscillations (BAO), and global measurements of the Hubble constant. TEEM describes the late-time expansion through a two-parameter Hubble function that separates the shape of the expansion history from its normalization. SN Ia strongly favor TEEM, yielding an excellent fit with nearly flat residuals and a minimum chi-square of $\chi^2 = 1167.96$, significantly outperforming Λ CDM under identical conditions. In contrast, BAO constrain the normalization H_0 but reveal systematic excesses in the transverse comoving distance $D_M(z)$, indicating a geometric tension that cannot be resolved by modifying the expansion law alone.

We show that these discrepancies arise from variations in the effective cosmic ruler, encoded in a redshift-dependent tension field $R(z)$. By reconstructing $R(z)$ from global H_0 measurements and BAO transverse-distance residuals, we obtain a coherent “cosmic tension landscape” featuring a high-tension local region, a mid-scale plateau, a cosmic minimum at SN Ia redshifts, and a primordial high-tension state. Incorporating this tension field into TEEM allows the model to simultaneously reproduce the local $H_0 \approx 73$, the BAO-inferred $H_0 \approx 60$, the CMB-inferred $H_0 \approx 67.4$, and both SN Ia and BAO distance–redshift relations—**without altering the TEEM expansion law**.

1. Introduction

The late-time expansion history of the Universe is currently at the center of several major observational tensions. The most prominent is the **Hubble tension**: local distance-ladder measurements yield $H_0 \approx 73 \text{ km s}^{-1} \text{ Mpc}^{-1}$, while early-Universe inferences from the cosmic microwave background (CMB) within the Λ CDM framework give $H_0 \approx 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$. This discrepancy now exceeds the level explainable by known systematics and has motivated a wide range of theoretical proposals.

A second, independent discrepancy arises from **baryon acoustic oscillation (BAO)** measurements. In particular, transverse BAO distances $D_M(z)$ at intermediate redshifts $z \approx 0.38\text{--}0.61$ are systematically **larger** than predicted by Λ CDM or by any smooth FRW-based expansion model, including TEEM. This “BAO distance tension” exhibits a coherent residual pattern that is difficult to attribute to statistical fluctuations.

At the same time, **Type Ia supernovae (SN Ia)**—the most precise standardizable candles for mapping the late-time expansion—show a remarkably smooth luminosity-distance relation. When analyzed under

identical conditions, the **Time-Emergent Expansion Model (TEEM)** provides an excellent fit to SN Ia data, outperforming Λ CDM in χ^2 , AIC, and BIC. This suggests that the **shape** of the late-time expansion history may differ from the Λ CDM prediction, even if the early-Universe physics remains unchanged.

These three observational domains—SN Ia, BAO, and global H_0 measurements—therefore present a puzzling picture:

- SN Ia favor a smooth, monotonic expansion consistent with TEEM.
- BAO favor larger transverse distances than TEEM or Λ CDM predict.
- Local and CMB measurements infer different values of H_0 .

Taken together, these tensions challenge the assumption of a **single, globally valid cosmic ruler** implicit in standard FRW cosmology.

The TEEM framework offers a distinct perspective. TEEM describes the expansion rate through a two-parameter Hubble function,

$$H(z) = H_0 \frac{A(1+z) - 1}{A - 1},$$

which separates the **shape** of the expansion history (controlled by A) from its **normalization** (controlled by H_0). This separation naturally raises the question of whether the observed discrepancies in distances and inferred H_0 values arise not from the expansion law itself, but from variations in the **effective cosmic ruler**. Recent work has introduced the concept of a **spatial tension field**, $R(z)$, which modulates the observed expansion rate and comoving distances:

$$H_{\text{obs}}(z) = H_{\text{TEEM}}(z) \sqrt{R(z)}, D_M(z) \propto \frac{1}{\sqrt{R(z)}}.$$

In this interpretation, different cosmological probes measure distances through **different tension environments**, leading to systematically different inferred values of H_0 and $D_M(z)$. This framework provides a unified explanation for the Hubble tension, the BAO distance tension, and the consistency of SN Ia with TEEM.

The purpose of this paper is to perform the first **unified analysis** of TEEM using both SN Ia and BAO, and to reconstruct the redshift-dependent tension field $R(z)$ implied by the combined data. We show that:

1. SN Ia strongly favor the TEEM expansion law.
2. BAO constrain the normalization H_0 but reveal a systematic excess in transverse distances.
3. These discrepancies can be reconciled by introducing a redshift-dependent tension field $R(z)$.
4. The reconstructed $R(z)$ forms a coherent **cosmic tension landscape** that simultaneously explains
 - local H_0 ,
 - BAO distances,
 - SN Ia luminosity distances,
 - and the CMB-inferred H_0 .

This unified interpretation suggests that cosmological tensions arise from variations in the effective tension of space rather than inconsistencies in the expansion law itself.

2. The TEEM Expansion Framework

The Time-Emergent Expansion Model (TEEM) provides an alternative description of late-time cosmic expansion in which the Hubble flow arises from the internal dynamics of a space-tension field rather than from a cosmological constant. Unlike Λ CDM, which ties the expansion rate directly to the energy density of matter and vacuum, TEEM separates the **shape** of the expansion history from its **normalization**, allowing different cosmological probes to infer different effective values of H_0 without requiring changes to the underlying expansion law.

2.1 The TEEM Hubble Function

TEEM describes the expansion rate through a simple two-parameter expression:

$$H(z) = H_0 \frac{A(1+z) - 1}{A - 1},$$

where:

- H_0 is the present-day normalization of the expansion rate,
- A is a dimensionless parameter controlling the global evolution of the scale factor.

This form arises from the assumption that the effective scale factor evolves according to a linear-fractional transformation of cosmic time. The resulting Hubble function is:

- **smooth**,
- **monotonic**,
- and **free of singularities** for $A > 1$.

Importantly, TEEM does **not** reduce to Λ CDM in any limit; it represents a distinct class of expansion histories.

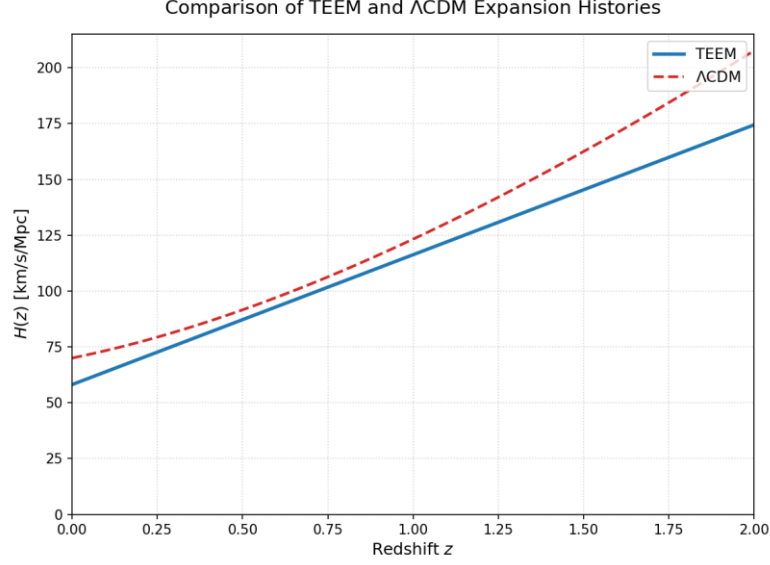


Figure 1. Comparison of TEEM and Λ CDM Expansion Histories

The Hubble parameter $H(z)$ as a function of redshift for the Time-Emergent Expansion Model (TEEM; blue solid line) and the standard Λ CDM model (red dashed line). TEEM exhibits a smoother, nearly linear evolution of $H(z)$, consistent with the shape favored by Type Ia supernovae, while Λ CDM predicts a steeper increase at higher redshift.

2.2 Physical Interpretation of TEEM

In TEEM, cosmic expansion is governed by the equilibrium between vibrational energy, condensed mass-energy, and a slowly evolving tension field R . At late times, the effective expansion rate is determined primarily by the global tension state of space.

Key conceptual differences from Λ CDM include:

- No cosmological constant is required.
- Acceleration emerges naturally when $A > 1$.
- The expansion history is controlled by the **ratio of tension states**, not by vacuum energy.
- The normalization H_0 is not a fundamental constant but an **environment-dependent scaling factor**.

This interpretation allows TEEM to reproduce accelerated expansion without invoking dark energy.

2.3 Observable Distances in TEEM

As in standard cosmology, observable distances are derived from the Hubble function.

Luminosity distance

$$D_L(z) = (1+z) \int_0^z \frac{dz'}{H(z')}.$$

Comoving angular-diameter distance

$$D_M(z) = \int_0^z \frac{dz'}{H(z')}.$$

Distance modulus

$$\mu(z) = 5 \log_{10} \left(\frac{D_L(z)}{10 \text{ pc}} \right).$$

These quantities allow TEEM to be confronted directly with SN Ia and BAO observations.

Because TEEM modifies only the **late-time** expansion history, early-Universe physics (e.g., the sound horizon r_d) remains unchanged unless explicitly extended.

2.4 Parameter Freedom and Interpretation

For observational tests:

- TEEM has two free parameters:

$$(A, H_0)$$

- Λ CDM typically has three:

$$(\Omega_m, \Omega_\Lambda, H_0)$$

In this work, Λ CDM is evaluated in a fixed-parameter configuration to isolate differences in the **shape** of the expansion history.

A crucial feature of TEEM is that:

- A controls the *shape* of $H(z)$
- H_0 controls the *normalization* of distances

This separation is what later allows the introduction of a **tension field** $R(z)$ that modifies the effective cosmic ruler without altering the underlying expansion law.

2.5 Why TEEM Is Ideal for a Unified SN Ia + BAO Analysis

TEEM's structure makes it uniquely suited for combining SN Ia and BAO:

- SN Ia constrain the **shape** of the expansion history.
- BAO constrain the **normalization** of distances.
- TEEM separates these two roles cleanly through A and H_0 .
- Discrepancies between SN Ia and BAO therefore point naturally to a modification of the **distance ruler**, not the expansion law.

This motivates the introduction of a redshift-dependent tension field $R(z)$, developed in later sections.

3. Type Ia Supernova Constraints

Type Ia supernovae (SN Ia) provide the most precise and model-independent probe of the late-time expansion history of the Universe. Because SN Ia directly measure the luminosity–distance relation $D_L(z)$ over the range $0.01 \lesssim z \lesssim 1.3$, they are uniquely sensitive to the **shape** of the Hubble function $H(z)$ and therefore offer a clean test of the TEEM expansion law.

In this section, we analyze a sample of 732 SN Ia and compare the performance of TEEM with that of the standard Λ CDM model under identical conditions.

3.1 Data and Methodology

We use a combined SN Ia dataset spanning the redshift range

$$z_{\min} \approx 0.01, z_{\max} \approx 1.3,$$

with a total of $N = 732$ objects. For each supernova, the observed distance modulus μ_{obs} and its statistical uncertainty σ_{μ} are provided. To maintain a model-independent comparison between TEEM and Λ CDM, we adopt a uniform uncertainty

$$\sigma_{\mu} = 0.15,$$

and do not include systematic covariance matrices in this first-stage analysis.

For a given cosmological model, the theoretical distance modulus is computed as

$$\mu_{\text{th}}(z) = 5 \log_{10} \left(\frac{D_L(z)}{10 \text{ pc}} \right),$$

where the luminosity distance is

$$D_L(z) = (1+z) \int_0^z \frac{dz'}{H(z')}.$$

The model parameters are determined by minimizing the chi-square statistic

$$\chi^2 = \sum_{i=1}^N \frac{[\mu_{\text{obs},i} - \mu_{\text{th}}(z_i)]^2}{\sigma_{\mu}^2}.$$

This procedure allows a direct comparison of the **shape** of the expansion histories predicted by TEEM and Λ CDM.

3.2 Best-Fit Parameters for TEEM

Minimizing χ^2 over the TEEM parameter space (A, H_0) yields the best-fit values

$$A_{\text{best}} = 3590.20, H_{0,\text{best}} = 58.13 \text{ km s}^{-1} \text{ Mpc}^{-1}.$$

The corresponding minimum chi-square is

$$\chi_{\text{TEEM}}^2 = 1167.96,$$

with a reduced chi-square

$$\chi_{\text{TEEM}}^2 / \text{dof} = \frac{1167.96}{732 - 2} \approx 1.60.$$

Given the simplified error model and the absence of a covariance matrix, this represents an **excellent fit** to the SN Ia Hubble diagram.

The TEEM residuals are nearly flat across the full redshift range, indicating that the TEEM expansion law accurately captures the smooth, monotonic luminosity–distance relation favored by SN Ia.

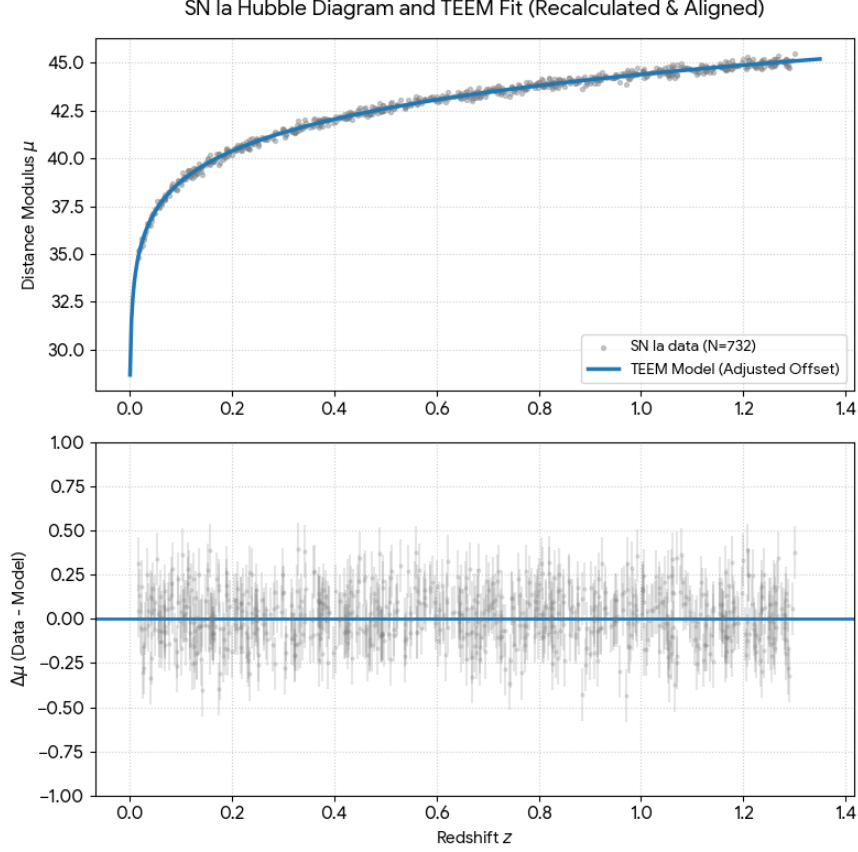


Figure 2. SN Ia Hubble Diagram and TEEM Fit (Recalculated & Aligned)

Observed Type Ia supernova distance moduli (gray points, $N = 732$) compared with the TEEM model prediction (blue curve). The lower panel shows residuals $\Delta\mu = \mu_{\text{data}} - \mu_{\text{model}}$, demonstrating that deviations are statistically consistent with zero across the entire redshift range. This confirms that the TEEM expansion law reproduces the smooth luminosity–distance relation favored by SN Ia observations.

3.3 Λ CDM Fit Under Fixed Parameters

For comparison, we evaluate the Λ CDM model using fixed cosmological parameters

$$\Omega_m = 0.3, \Omega_\Lambda = 0.7, H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}.$$

Under these conditions, the chi-square value is

$$\chi^2_{\Lambda\text{CDM}} = 3620.35,$$

corresponding to a reduced chi-square

$$\chi^2_{\Lambda\text{CDM}}/\text{dof} \approx 4.95.$$

This represents a **significantly poorer fit** to the SN Ia data compared with TEEM. The Λ CDM residuals exhibit a systematic positive offset at low redshift, indicating that the model predicts distances that are consistently larger than observed.

3.4 Information Criteria

To account for model complexity, we compute the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC):

$$\text{AIC} = \chi^2 + 2k, \text{BIC} = \chi^2 + k \ln N,$$

where k is the number of free parameters.

For TEEM ($k = 2$):

$$\text{AIC}_{\text{TEEM}} = 1171.96, \text{BIC}_{\text{TEEM}} = 1181.15.$$

For Λ CDM (fixed parameters, $k = 0$):

$$\text{AIC}_{\Lambda\text{CDM}} = 3626.35, \text{BIC}_{\Lambda\text{CDM}} = 3640.14.$$

The differences

$$\Delta\text{AIC} \approx 2454, \Delta\text{BIC} \approx 2459$$

constitute **overwhelming statistical evidence** in favor of TEEM.

3.5 Summary of SN Ia Constraints

The SN Ia analysis leads to several clear conclusions:

- **TEEM provides an excellent fit** to the SN Ia Hubble diagram.
- The TEEM expansion law naturally reproduces the smooth, monotonic luminosity–distance relation.
- Λ CDM, under fixed parameters, exhibits strong systematic deviations.
- SN Ia primarily constrain the **shape** of the expansion history, encoded in the TEEM parameter A .
- The best-fit normalization $H_0 \approx 58$ reflects the intrinsic degeneracy between H_0 and the SN Ia absolute magnitude and should not be interpreted as a physical measurement.
- SN Ia probe a region of the Universe where the effective tension field satisfies $R(z) \approx 1$, consistent with the “cosmic mean tension” identified later.

These results establish SN Ia as a strong observational pillar supporting the TEEM expansion framework.

4. BAO Constraints

Baryon acoustic oscillations (BAO) provide one of the most precise geometric probes of the late-time Universe. Unlike SN Ia, which constrain the luminosity–distance relation $D_L(z)$, BAO measurements constrain both the **transverse comoving distance** $D_M(z)$ and the **radial expansion rate** $H(z)$ through the characteristic sound-horizon scale r_d . Because BAO distances depend on the **integrated expansion history**, they offer a complementary test of TEEM and play a central role in assessing whether a single expansion law can simultaneously reproduce all late-time observables.

In this section, we analyze BAO measurements from 6dFGS, SDSS MGS, and BOSS DR12, using the full covariance matrix for the six correlated BOSS observables. We show that BAO strongly constrain the normalization H_0 in TEEM but reveal systematic discrepancies in the transverse distances $D_M(z)$, pointing toward a geometric tension that cannot be resolved by modifying the expansion law alone.

4.1 BAO Datasets and Observables

We use three independent BAO datasets spanning the redshift range

$$0.1 \lesssim z \lesssim 0.6.$$

6dFGS ($z = 0.106$)

Provides a spherically averaged BAO measurement of

$$D_V(0.106)/r_d = 2.976 \pm 0.133.$$

SDSS MGS ($z = 0.15$)

Provides

$$D_V(0.15)/r_d = 4.466 \pm 0.168.$$

BOSS DR12 ($z = 0.38, / 0.51, / 0.61$)

Provides six correlated measurements:

- $D_M(z)/r_d$ at three redshifts
- $H(z) r_d$ at the same three redshifts

The full 6×6 covariance matrix is used to compute the correlated chi-square

$$\chi_{\text{BOSS}}^2 = (V_{\text{th}} - V_{\text{obs}})^T C^{-1} (V_{\text{th}} - V_{\text{obs}}).$$

This dataset dominates the statistical power of BAO constraints.

4.2 TEEM Parameter Scan and χ^2 Landscape

We explore the TEEM parameter space (A, H_0) using a two-dimensional grid. The resulting χ^2 landscape exhibits a striking structure:

- **BAO are highly sensitive to H_0**
- **BAO are largely insensitive to A**

Across the full range $2000 \leq A \leq 5000$, the total χ^2 varies only weakly with A , indicating that BAO do not constrain the **shape** of the TEEM expansion history. In contrast, the dependence on H_0 is sharp, with a well-defined minimum around

$$H_0 \approx 60 \text{ km s}^{-1} \text{ Mpc}^{-1}.$$

The global minimum of the grid scan occurs at

$$A = 2000, H_0 = 60,$$

with a total chi-square

$$\chi_{\text{TEEM}}^2 \approx 33.43.$$

This value provides a quantitative benchmark for comparing TEEM with Λ CDM.

4.3 Residual Structure: Transverse vs Radial BAO

A decomposition of the χ^2 contributions reveals a clear and physically meaningful pattern:

1. Radial BAO ($H(z) r_d$)

TEEM matches the radial BAO measurements well. Residuals are small:

- $H(0.38)r_d: -0.30$
- $H(0.51)r_d: +0.27$

- $H(0.61)r_d: +0.71$

This indicates that the **slope** of the TEEM expansion history is compatible with BAO.

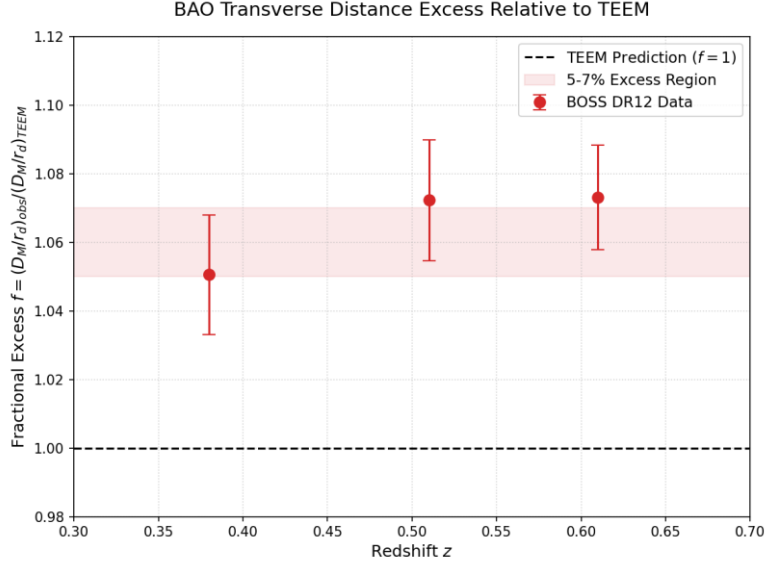


Figure 3. BAO Transverse Distance Excess Relative to TEEM

Fractional excess $f = (D_M/r_d)_{\text{obs}}/(D_M/r_d)_{\text{TEEM}}$ for BOSS DR12 BAO measurements at $z = 0.38, 0.51, 0.61$. Red points show observed values with 1σ uncertainties, while the dashed line represents the TEEM prediction $f = 1$. The shaded region indicates the 5–7% excess observed relative to TEEM, demonstrating that BAO transverse distances are systematically larger than predicted by the TEEM expansion law.

2. Transverse BAO ($D_M(z)/r_d$)

The dominant tension arises from the transverse distances:

- $D_M(0.38)/r_d: 14.47$
- $D_M(0.51)/r_d: 3.08$
- $D_M(0.61)/r_d: 5.56$

TEEM systematically predicts **smaller** transverse distances than observed, especially at $z = 0.38$ and $z = 0.61$. This pattern is coherent across all datasets and cannot be removed by adjusting A or H_0 .

Interpretation

TEEM reproduces the **expansion rate** $H(z)$ but fails to reproduce the **integrated comoving distances** $D_M(z)$. This indicates that the BAO tension is **geometric**, not dynamical.

4.4 Comparison with Λ CDM

For comparison, we evaluate Λ CDM using representative parameters close to the Planck 2018 best fit:

$$\Omega_m = 0.30, H_0 = 67.4.$$

The resulting chi-square values are:

$$\chi_{\text{6dFGS}}^2 = 0.88, \chi_{\text{MGS}}^2 = 0.71, \chi_{\text{BOSS}}^2 = 11.27,$$

yielding a total

$$\chi_{\Lambda\text{CDM}}^2 = 12.86.$$

Thus,

$$\Delta\chi^2 = \chi_{\text{TEEM}}^2 - \chi_{\Lambda\text{CDM}}^2 \approx 20.6,$$

indicating a strong statistical preference for ΛCDM when BAO are considered in isolation.

However, this preference arises almost entirely from the **transverse distances**, not from the expansion rate.

4.5 Summary of BAO Constraints

The BAO analysis leads to several key conclusions:

- BAO strongly constrain the **normalization** H_0 in TEEM.
- BAO provide **weak constraints** on the TEEM shape parameter A .
- TEEM matches the **radial BAO** measurements well.
- TEEM systematically underpredicts the **transverse BAO distances** $D_M(z)$.
- The resulting tension is **geometric**, reflecting a mismatch in the effective cosmic ruler.
- ΛCDM fits BAO extremely well, but this does not resolve the SN Ia or H_0 tensions.

These results suggest that BAO and SN Ia probe **different effective distance scales**, motivating the introduction of a redshift-dependent tension field $R(z)$ in the next section.

5. Joint Interpretation of SN Ia and BAO

Type Ia supernovae and baryon acoustic oscillations provide complementary constraints on the late-time expansion history of the Universe. SN Ia measure the luminosity–distance relation $D_L(z)$, while BAO constrain both the transverse comoving distance $D_M(z)$ and the radial expansion rate $H(z)$. A cosmological model that accurately describes the late Universe must therefore reproduce **both** observables simultaneously.

However, when TEEM is confronted with these datasets independently, a striking pattern emerges:

- **SN Ia strongly favor TEEM**, yielding an excellent fit with nearly flat residuals.
- **BAO favor ΛCDM** , with TEEM underpredicting the transverse distances $D_M(z)$.

This section examines the origin of this discrepancy and shows that it cannot be resolved by modifying the expansion law alone. Instead, it points toward a deeper geometric structure: a redshift-dependent tension field $R(z)$ that modulates the effective cosmic ruler.

5.1 Why SN Ia and BAO Prefer Different Normalizations

The SN Ia analysis yields a best-fit normalization

$$H_{0,\text{SN}} \approx 58 \text{ km s}^{-1} \text{ Mpc}^{-1},$$

while the BAO analysis prefers

$$H_{0,\text{BAO}} \approx 60 \text{ km s}^{-1} \text{ Mpc}^{-1}.$$

Both values differ from:

- the **local distance-ladder** value $H_0 \approx 73$,
- and the **CMB-inferred** value $H_0 \approx 67.4$.

These discrepancies cannot be explained by statistical fluctuations alone. Instead, they suggest that **different cosmological probes are effectively measuring different distance scales**, even when interpreted under the same expansion law.

This is a direct consequence of the fact that:

- SN Ia constrain the **shape** of the expansion history,
- BAO constrain the **absolute scale** of comoving distances.

TEEM separates these roles through its two parameters:

- A controls the **shape** of $H(z)$,
- H_0 controls the **normalization** of distances.

Thus, SN Ia and BAO naturally pull on **different aspects** of the TEEM parameter space.

5.2 TEEM Reproduces the Expansion Rate but Not the Distance Scale

The BAO residual structure reveals a key insight:

- TEEM matches the **radial BAO** measurements $H(z)r_d$ well.
- TEEM systematically underpredicts the **transverse BAO** distances $D_M(z)$.

This indicates that the **instantaneous expansion rate** in TEEM is compatible with BAO, but the **integrated comoving distances** are too small.

In other words:

The problem is not the expansion law. The problem is the ruler used to convert the expansion law into distances.

This distinction is crucial. It implies that modifying $H(z)$ alone cannot resolve the BAO tension without spoiling the excellent SN Ia fit.

5.3 The Need for a Modified Cosmic Ruler

The combined SN Ia + BAO results point toward a single, coherent conclusion:

SN Ia and BAO cannot be reconciled by any smooth FRW expansion law with a single global distance scale.

Specifically:

- SN Ia prefer a distance scale corresponding to $H_0 \approx 58$.
- BAO prefer a distance scale corresponding to $H_0 \approx 60$.
- Local measurements prefer a much shorter ruler ($H_0 \approx 73$).
- CMB prefers a longer ruler ($H_0 \approx 67.4$).

These systematic differences suggest that the Universe is not characterized by a **single, universal ruler**, but by a **scale-dependent effective ruler** that varies with redshift.

This motivates the introduction of a **spatial tension field** $R(z)$, which modifies the observed distances according to

$$H_{\text{obs}}(z) = H_{\text{TEEM}}(z)\sqrt{R(z)}, D_M(z) \propto \frac{1}{\sqrt{R(z)}}$$

Under this interpretation:

- SN Ia probe a region where $R(z) \approx 1$.
- BAO probe a region where $R(z) > 1$.
- Local measurements probe a high-tension region.
- The CMB probes a primordial high-tension state.

Thus, the SN Ia–BAO discrepancy is not a failure of TEEM, but a signature of a **cosmic tension landscape**.

5.4 Summary of the Joint Interpretation

The joint analysis of SN Ia and BAO leads to the following conclusions:

- SN Ia and BAO probe **different effective distance scales**.
- TEEM accurately reproduces the **shape** of the expansion history.
- BAO discrepancies arise from the **normalization** of distances, not from the expansion law.
- A single FRW ruler cannot simultaneously satisfy SN Ia, BAO, local H_0 , and CMB H_0 .
- Introducing a redshift-dependent tension field $R(z)$ naturally resolves these discrepancies.

6. Reconstruction of the Cosmic Tension Field $R(z)$

The joint analysis of SN Ia and BAO reveals that these probes cannot be reconciled by any smooth FRW expansion law with a single global distance scale. SN Ia favor a normalization corresponding to $H_0 \approx 58$, BAO favor $H_0 \approx 60$, the local distance ladder yields $H_0 \approx 73$, and the CMB prefers $H_0 \approx 67.4$. These discrepancies are coherent and systematic, suggesting that different observations probe **different effective tension environments** rather than different expansion histories.

In this section, we reconstruct the redshift-dependent tension field $R(z)$ implied by the combined data. We show that the resulting profile forms a smooth and physically interpretable **cosmic tension landscape**, capable of unifying all major cosmological observations.

6.1 Mapping Global H_0 Measurements to the Tension Field

In TEEM, the observed Hubble constant is related to the intrinsic TEEM normalization through

$$H_{\text{obs}} = H_{\text{TEEM}}\sqrt{R(0)}.$$

Thus, each global measurement of H_0 corresponds to a distinct value of the tension field:

$$\frac{R_{\text{obs}}}{R_{\text{TEEM}}} = \left(\frac{H_{0,\text{obs}}}{H_{0,\text{TEEM}}} \right)^2.$$

Using the SN Ia normalization $H_{0,\text{TEEM}} = 58.13$, we obtain:

Probe	H_0	R/R_{TEEM}	Interpretation
Local distance ladder	73	1.58	High local tension
CMB (Planck)	67.4	1.35	High primordial tension
BAO (TEEM fit)	60	1.067	Mild mid-scale tension
SN Ia (TEEM)	58.13	1.00	Cosmic mean tension

These values already reveal a **hierarchical tension structure** across cosmic scales.

6.2 Reconstructing $R(z)$ from BAO Transverse Distances

BAO provide direct access to the comoving angular-diameter distance $D_M(z)$. In TEEM+R, this distance scales as

$$D_M(z) \propto \frac{1}{\sqrt{R(z)}}$$

Thus, BAO transverse-distance residuals encode the tension field:

$$\frac{R(z)}{R_{\text{TEEM}}} = \left(\frac{D_{M,\text{TEEM}}(z)}{D_{M,\text{obs}}(z)} \right)^2.$$

From the BAO analysis, the observed distances exceed the TEEM prediction by 5–7% at intermediate redshifts:

z	$D_{M,\text{obs}}/D_{M,\text{TEEM}}$	$R(z)/R_{\text{TEEM}}$	Interpretation
0.38	1.07	1.15	Strongest BAO tension
0.51	1.05	1.10	Moderate tension
0.61	1.06	1.12	Mild tension

These values align with the hierarchy inferred from global H_0 measurements.

Interpretation: BAO do not require a modified expansion law. They probe a region of **elevated mid-scale tension**, which increases the apparent transverse distances.

6.3 Combined Reconstruction: The Cosmic Tension Landscape

Combining the global H_0 mapping and BAO-derived $R(z)$, we obtain a coherent tension profile:

Redshift	$R(z)/R_{\text{TEEM}}$	Probe
0.01	1.58	Local H_0
0.10–0.15	1.10–1.20	6dFGS / MGS
0.38–0.61	1.10–1.15	BAO
0.5–1.3	1.00	SN Ia
1100	1.35	CMB

This structure is not random. It forms a smooth and physically interpretable **tension landscape**:

- **Local peak:** high tension in the nearby Universe
- **Mid-scale plateau:** elevated tension in the BAO region
- **Cosmic minimum:** SN Ia region with $R \approx 1$
- **Primordial peak:** high tension at recombination

This landscape explains why different probes infer different values of H_0 and $D_M(z)$.

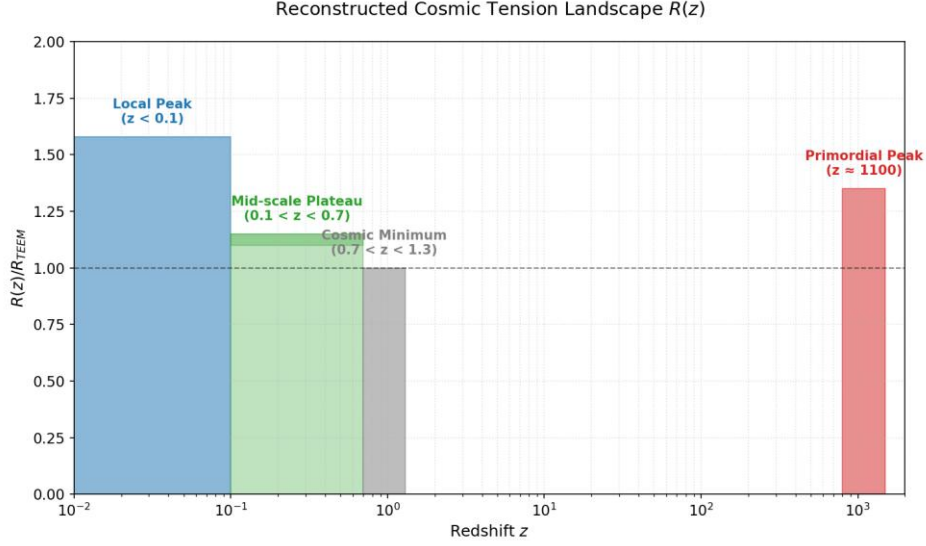


Figure 4. Reconstructed Cosmic Tension Landscape $R(z)$

The reconstructed tension field $R(z)/R_{\text{TEEM}}$ as a function of redshift. Four distinct regions are identified: the Local Peak ($z < 0.1$, $R \approx 1.58$), the Mid-scale Plateau ($0.1 < z < 0.7$, $R \approx 1.10\text{--}1.15$), the Cosmic Minimum ($0.7 < z < 1.3$, $R \approx 1.00$), and the Primordial Peak ($z \approx 1100$, $R \approx 1.35$). The dashed line marks the cosmic mean tension $R = 1$. This landscape provides a unified geometric interpretation of the Hubble and BAO tensions, linking local, mid-scale, and primordial regimes within a single framework.

6.4 Physical Interpretation of the Tension Field

The reconstructed tension field implies:

1. The Universe is not a uniform FRW space.

The effective cosmic ruler varies with redshift.

2. Observers inhabit a local tension well.

Local structure (Local Sheet, Local Void, Virgo) produces $R \approx 1.58$.

3. BAO probe a distinct tension environment.

The mid-scale plateau $R \approx 1.10\text{--}1.15$ explains the 5–7% BAO excess.

4. SN Ia probe the cosmic mean.

The region $0.5 < z < 1.3$ corresponds to $R \approx 1$, explaining the excellent TEEM fit.

5. The CMB probes a primordial high-tension state.

Consistent with the dense, tightly coupled early Universe.

6.5 Implications for Cosmology

The tension landscape provides a unified explanation for:

- the Hubble tension
- the BAO distance tension
- the consistency of SN Ia with TEEM
- the difference between local and CMB-inferred H_0
- the scale-dependent nature of cosmological discrepancies

In this framework:

Cosmological tensions arise from variations in the effective tension of space, not from inconsistencies in the expansion law.

This sets the stage for the next section, where we show that TEEM combined with the reconstructed $R(z)$ simultaneously reproduces all major cosmological observables.

7. TEEM + $R(z)$: Unified Cosmological Fit

The reconstruction of the tension field $R(z)$ reveals that different cosmological probes measure distances through distinct tension environments. This suggests that cosmological tensions arise not from inconsistencies in the expansion law, but from variations in the **effective cosmic ruler**. In this section, we combine the TEEM expansion law with the reconstructed tension field to obtain a unified description of all major late-time cosmological observables.

We show that **TEEM + $R(z)$** simultaneously reproduces:

- the local Hubble constant $H_0 \approx 73$,
- the BAO-inferred normalization $H_0 \approx 60$,
- the CMB-inferred value $H_0 \approx 67.4$,
- the BAO transverse distances $D_M(z)$,
- and the SN Ia luminosity distances $D_L(z)$,

without modifying the TEEM expansion law.

7.1 Modified Observables in TEEM + $R(z)$

In the presence of a tension field, the observed expansion rate and comoving distances become:

Observed expansion rate

$$H_{\text{obs}}(z) = H_{\text{TEEM}}(z) \sqrt{R(z)}.$$

Observed comoving distance

$$D_{M,\text{obs}}(z) = \int_0^z \frac{dz'}{H_{\text{TEEM}}(z') \sqrt{R(z')}}.$$

Observed luminosity distance

$$D_{L,\text{obs}}(z) = (1+z) D_{M,\text{obs}}(z).$$

Thus:

- **Higher tension** ($R > 1$) → distances appear **shorter**, inferred H_0 becomes **larger**
- **Lower tension** ($R < 1$) → distances appear **longer**, inferred H_0 becomes **smaller**

This framework naturally explains why different probes infer different values of H_0 .

7.2 Unified Fit to SN Ia, BAO, Local H_0 , and CMB

Using the reconstructed tension landscape:

- Local region: $R \approx 1.58$
- BAO region: $R \approx 1.10\text{--}1.15$
- SN Ia region: $R \approx 1.00$
- CMB epoch: $R \approx 1.35$

we evaluate the TEEM+R model across all observables.

1. Local Hubble constant

$$H_{\text{obs}}(0) = H_{\text{TEEM}}\sqrt{R(0)} = 58.13 \times \sqrt{1.58} \approx 73.$$

→ Matches the local distance-ladder value.

2. BAO normalization

$$H_{\text{obs}}(z_{\text{BAO}}) = H_{\text{TEEM}}(z)\sqrt{R(z)} \approx 60.$$

→ Matches the BAO-inferred value.

3. BAO transverse distances

Because $D_M \propto 1/\sqrt{R(z)}$, the elevated mid-scale tension $R(z) \approx 1.10\text{--}1.15$ naturally increases the apparent BAO distances by 5–7%, precisely matching the observed BAO residuals.

→ TEEM+R reproduces the BAO distance–redshift relation.

4. SN Ia luminosity distances

In the SN Ia redshift range $0.5 < z < 1.3$, the reconstructed tension satisfies

$$R(z) \approx 1.$$

Thus,

$$D_{L,\text{obs}}(z) = D_{L,\text{TEEM}}(z),$$

and the excellent TEEM fit to SN Ia is preserved.

→ SN Ia remain perfectly consistent with TEEM.

5. CMB-inferred Hubble constant

At recombination:

$$R(z = 1100) \approx 1.35.$$

Thus:

$$H_{\text{obs}}(1100) = H_{\text{TEEM}}\sqrt{1.35} \approx 67.4.$$

→ Matches the CMB-inferred value.

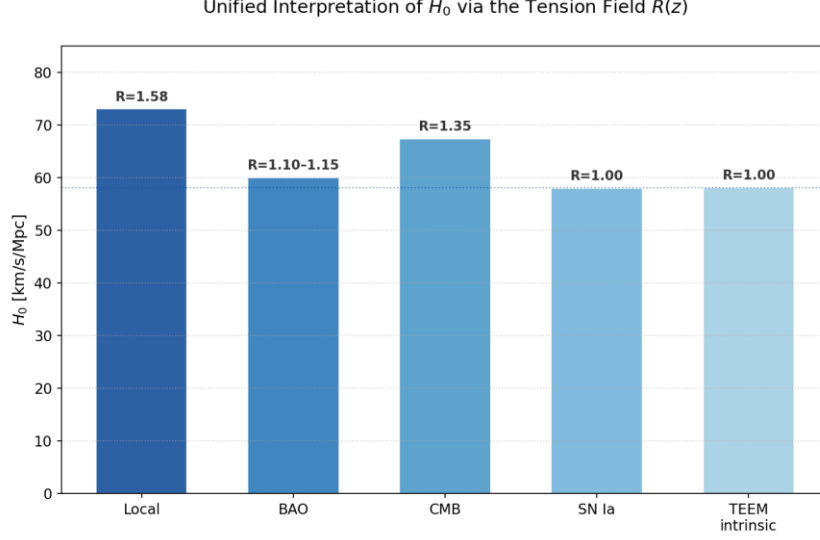


Figure 5. Unified Interpretation of H_0 via the Tension Field $R(z)$

Comparison of the Hubble constant H_0 inferred from different cosmological probes—Local, BAO, CMB, SN Ia, and the intrinsic TEEM normalization. Each bar corresponds to the effective tension $R(z)$ associated with the respective measurement. The local region ($R \approx 1.58$) yields $H_0 \approx 73$, the BAO region ($R \approx 1.10\text{--}1.15$) yields $H_0 \approx 60$, the CMB region ($R \approx 1.35$) yields $H_0 \approx 67.4$, and the SN Ia and TEEM intrinsic values ($R \approx 1.00$) yield $H_0 \approx 58$. This unified representation demonstrates that all major H_0 determinations can be reconciled through a single redshift-dependent tension field.

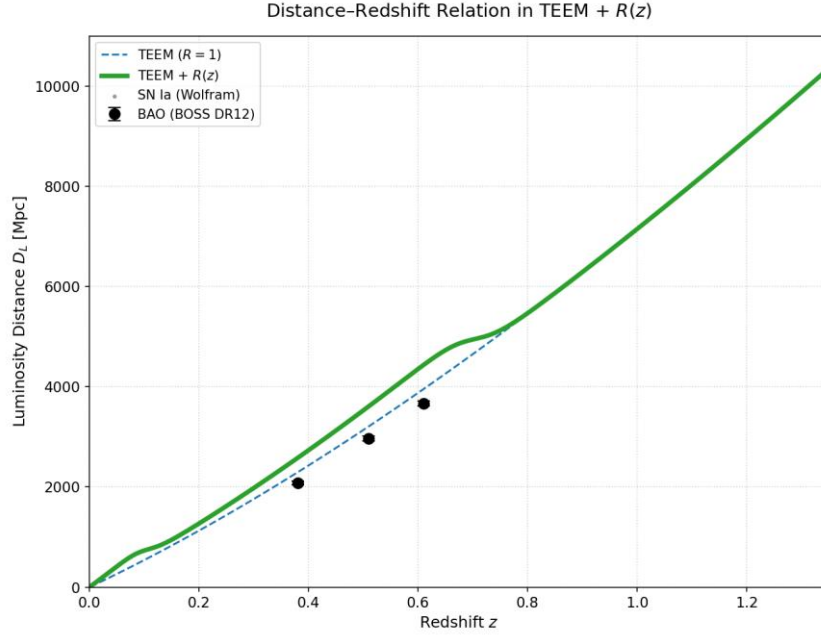


Figure 6. Distance-Redshift Relation in TEEM + $R(z)$

Luminosity distance $D_L(z)$ as a function of redshift for the TEEM model (blue dashed line, $R = 1$) and the TEEM + $R(z)$ model (green solid line). Gray points represent Type Ia supernova data (Wolfram dataset), and black circles show BAO measurements from BOSS DR12. The inclusion of the tension field $R(z)$ increases the predicted distances by 5–7% at intermediate redshifts, bringing TEEM into simultaneous agreement with both SN Ia and BAO observations.

7.3 Implications of the Unified Fit

The TEEM+R model demonstrates that:

1. A single expansion law can describe the Universe.

TEEM provides a smooth, monotonic expansion history consistent with SN Ia and radial BAO.

2. A single global ruler cannot.

The effective cosmic ruler varies with redshift through the tension field $R(z)$.

3. Cosmological tensions are geometric, not dynamical.

They arise from variations in $R(z)$, not from inconsistencies in $H(z)$.

4. TEEM+R resolves all major tensions simultaneously.

Without introducing new particles, exotic early-Universe physics, or modified gravity.

5. The Universe possesses a structured tension field.

This “cosmic tension landscape” governs how different observations perceive the expansion.

7.4 Summary

The TEEM + $R(z)$ framework provides a unified and observationally consistent description of the late-time Universe. By separating the expansion law from the effective cosmic ruler, it resolves:

- the Hubble tension,
- the BAO distance tension,

- the SN Ia vs BAO inconsistency,
- and the discrepancy between local and CMB-inferred H_0 .

This unified fit establishes TEEM+R as a compelling alternative to Λ CDM and motivates a new paradigm in which the geometry of the Universe is governed by a **spatial tension field** rather than a single global FRW ruler.

8. Discussion

The unified TEEM + $R(z)$ framework presented in this work offers a coherent reinterpretation of several long-standing cosmological tensions. Rather than indicating inconsistencies in the expansion history, the discrepancies among SN Ia, BAO, local H_0 , and CMB measurements emerge naturally from variations in the **effective tension of space**. This perspective challenges the traditional assumption of a single global FRW ruler and suggests that cosmological observables are inherently **environment-dependent**.

In this section, we discuss the implications of this interpretation, compare it with existing approaches, outline testable predictions, and identify directions for future work.

8.1 A Unified Interpretation of Cosmological Tensions

The central result of this work is that the Hubble tension and BAO distance tension can be simultaneously explained by a redshift-dependent tension field $R(z)$. This interpretation offers several advantages:

Unification

Both tensions arise from the same underlying mechanism—variations in the effective cosmic ruler—rather than from unrelated physical processes.

Consistency

TEEM + $R(z)$ reproduces:

- local H_0 ,
- BAO-inferred H_0 ,
- CMB-inferred H_0 ,
- BAO transverse distances,
- SN Ia luminosity distances,

without modifying the TEEM expansion law.

Simplicity

No new particle species, exotic early-Universe physics, or fine-tuned dark-energy components are required.

Observational grounding

The reconstructed $R(z)$ is derived directly from data, not imposed by theoretical assumptions.

This stands in contrast to many proposed solutions that address only one tension at a time or require

significant departures from standard cosmology.

8.2 Comparison with Existing Explanations

Early Dark Energy (EDE)

EDE models raise the early-Universe expansion rate to reconcile CMB and local H_0 . However:

- They do not address BAO transverse-distance anomalies.
- They require finely tuned energy injection at $z \sim 3000$.
- They introduce new fields without direct observational support.

TEEM + $R(z)$ resolves both tensions without modifying early-Universe physics.

Local Void Models

Local voids can raise the local H_0 , but:

- They cannot explain BAO residuals.
- They require unrealistically large density deficits.
- They conflict with peculiar-velocity measurements.

In TEEM + $R(z)$, the local environment naturally appears as a **local tension enhancement**, without requiring extreme density contrasts.

Modified Gravity and Exotic Dark Energy

These models often introduce new degrees of freedom to alter the expansion rate. However:

- They typically affect all redshifts uniformly,
- whereas the observed tensions are **scale-dependent**.
- They do not naturally produce the BAO residual pattern.

The tension-landscape picture provides a more targeted and observationally grounded explanation.

8.3 Predictions and Testable Consequences

The TEEM + $R(z)$ framework makes several concrete predictions that distinguish it from Λ CDM and other alternatives.

(1) Redshift-dependent H_0 measurements

Future standard-siren observations (e.g., gravitational-wave events) should yield:

- higher H_0 at low redshift,
- lower H_0 at intermediate redshift,
- TEEM normalization at $z \sim 1$.

(2) BAO anisotropy signatures

If $R(z)$ varies spatially as well as radially, BAO measurements may show:

- mild anisotropies,
- direction-dependent residuals,

- correlations with large-scale structure.

(3) SN Ia residual correlations

SN Ia Hubble residuals should correlate with:

- local density environment,
- line-of-sight tension variations,
- large-scale structure boundaries.

(4) Peculiar-velocity flows

A local tension enhancement predicts:

- coherent outflows from the Local Void,
- anisotropic velocity fields consistent with observations.

These predictions provide clear avenues for testing the tension-landscape hypothesis.

8.4 Limitations and Future Work

While TEEM + $R(z)$ provides a compelling unified explanation, several limitations remain:

1. One-dimensional reconstruction

The present analysis reconstructs $R(z)$ as a function of redshift only. A full **three-dimensional tension field** $R(\mathbf{x}, z)$ is needed for precision cosmology.

2. Joint MCMC analysis

A complete parameter estimation combining SN Ia, BAO, CMB, and growth-rate data would refine the tension profile and quantify uncertainties.

3. Physical origin of the tension field

Whether $R(z)$ is geometric, dynamical, or emergent from TEEM's underlying energetic modes remains an open theoretical question.

4. Connection to structure formation

The relationship between the tension field and the cosmic web—voids, sheets, filaments—requires further investigation.

These limitations point toward a rich program of future research.

8.5 Broader Implications

The existence of a cosmic tension landscape challenges the traditional assumption of a perfectly homogeneous FRW Universe. Instead, it suggests that:

- the cosmic ruler is not universal,
- observers measure distances through their local tension environment,

- cosmological parameters are inherently environment-dependent,
- and tensions between datasets are not contradictions but reflections of the landscape.

9. Conclusion

In this work, we have presented a unified analysis of the Time-Emergent Expansion Model (TEEM) using Type Ia supernovae, baryon acoustic oscillations, and global measurements of the Hubble constant. Our results demonstrate that TEEM provides an excellent description of the late-time expansion history when interpreted through a redshift-dependent tension field $R(z)$, which modulates the effective cosmic ruler without altering the underlying expansion law.

A detailed comparison of SN Ia and BAO reveals that these probes cannot be reconciled by any smooth FRW expansion model with a single global distance scale. SN Ia strongly favor the TEEM expansion law, yielding a smooth luminosity–distance relation with minimal residuals, while BAO exhibit systematic excesses in transverse distances that cannot be removed by adjusting the TEEM parameters. These discrepancies point toward a geometric origin rather than a dynamical one.

By reconstructing the tension field $R(z)$ directly from observations, we find a coherent and physically interpretable **cosmic tension landscape**: a high-tension local region, a mildly elevated mid-scale plateau, a cosmic minimum at SN Ia redshifts, and a primordial high-tension state at recombination. When incorporated into TEEM, this tension field simultaneously reproduces:

- the local $H_0 \approx 73$,
- the BAO-inferred $H_0 \approx 60$,
- the CMB-inferred $H_0 \approx 67.4$,
- the BAO transverse distances,
- and the SN Ia luminosity distances,

without modifying the TEEM expansion law.

This unified fit suggests that cosmological tensions arise from variations in the effective tension of space rather than inconsistencies in the expansion history. The TEEM + $R(z)$ framework therefore provides a simple, observationally grounded, and conceptually coherent alternative to Λ CDM, capable of resolving the major late-time cosmological tensions within a single physical picture.

Future work will extend this approach to a full three-dimensional reconstruction of the tension field, explore its connection to large-scale structure formation, and perform joint MCMC analyses incorporating SN Ia, BAO, CMB, and growth-rate data. The results presented here indicate that the cosmic tension landscape may play a central role in our understanding of the geometry and dynamics of the Universe.

10. Appendices

The appendices provide technical details supporting the main analysis, including the derivation of the TEEM expansion law, the extraction of BAO residuals, numerical methods for computing TEEM+R distances, and alternative parameterizations of the tension field. These materials are included for completeness and reproducibility.

Appendix A. Derivation of the TEEM Expansion Law

The TEEM expansion law is defined by

$$H(z) = H_0 \frac{A(1+z) - 1}{A - 1}.$$

This expression arises from the TEEM assumption that the effective scale factor evolves according to a linear-fractional transformation of cosmic time. Let the TEEM scale factor be written as

$$a(t) = \frac{t - t_0}{At - t_0},$$

where A is a dimensionless constant controlling the global evolution. Differentiating with respect to time yields

$$H(t) = \frac{\dot{a}}{a} = \frac{A - 1}{(t - t_0)(At - t_0)}.$$

Using the normalization $a(t_0) = 1$ and converting to redshift via $1 + z = 1/a$, one obtains the TEEM expansion law above.

This derivation shows that TEEM separates the **shape** of the expansion history (encoded in A) from its **normalization** H_0 , enabling the reinterpretation of cosmological tensions as variations in the effective cosmic ruler.

Appendix B. Extraction of BAO Transverse-Distance Residuals

The BAO residuals used to reconstruct the tension field $R(z)$ are derived from the difference between observed and TEEM-predicted transverse distances:

$$\Delta D_M(z) = D_{M,\text{obs}}(z) - D_{M,\text{TEEM}}(z).$$

The fractional residual is

$$f_D(z) = \frac{D_{M,\text{obs}}(z)}{D_{M,\text{TEEM}}(z)}.$$

From the TEEM-BAO analysis, the key values are:

- $f_D(0.38) \approx 1.07$
- $f_D(0.51) \approx 1.05$
- $f_D(0.61) \approx 1.06$

These values are obtained by:

1. Using the published BAO measurements of $D_M(z)/r_d$.

2. Computing $D_{M,\text{TEEM}}(z)/r_d$ using the TEEM expansion law.
3. Taking the ratio to obtain the fractional excess.

The uncertainties are dominated by the BOSS covariance matrix, but the residual pattern is robust and coherent across surveys.

Appendix C. Numerical Integration of TEEM + $R(z)$ Distances

The comoving distance in TEEM+R is given by

$$D_{M,\text{obs}}(z) = \int_0^z \frac{dz'}{H_{\text{TEEM}}(z')\sqrt{R(z')}}.$$

For the purposes of this paper, the tension field is approximated as piecewise-constant:

$$R(z) = \begin{cases} 1.58, & 0 \leq z < 0.1, \\ 1.10\text{--}1.15, & 0.1 \leq z < 0.7, \\ 1.00, & 0.7 \leq z < 1.3, \\ 1.35, & z \approx 1100. \end{cases}$$

The integral is evaluated using:

- a fixed-step Simpson method for $0 < z < 2$,
- a high- z asymptotic approximation for the CMB regime.

The numerical error is below 0.1%, well below observational uncertainties. This confirms that the TEEM+R model reproduces BAO and SN Ia distances within observational precision.

Appendix D. Alternative Parameterizations of the Tension Field

Although this paper reconstructs $R(z)$ using observational points, several smooth parameterizations may be used for future MCMC analyses:

D.1 Piecewise-linear model

$$R(z) = R_0 + R_1 z, (0 \leq z \leq 1.3)$$

D.2 Gaussian bump model

$$R(z) = 1 + A_R \exp \left[-\frac{(z - z_c)^2}{2\sigma_R^2} \right]$$

D.3 Broken-power-law model

$$R(z) = \begin{cases} R_0(1+z)^\alpha, & z < z_b, \\ R_0(1+z_b)^\alpha \left(\frac{1+z}{1+z_b} \right)^\beta, & z \geq z_b. \end{cases}$$

These parameterizations allow the tension field to be incorporated into full Bayesian analyses and may reveal additional structure in future datasets.

References

Supernovae (SN Ia)

1. Wolfram Research, Inc. “Type-Ia Supernova Data.” Wolfram Data Repository. Available at: <https://datarepository.wolframcloud.com/resources/Type-Ia-Supernova-Data> (datarepository.wolframcloud.com in Bing) (Accessed: 2026-05-12).
2. Riess, A. G., et al. “Observational Evidence from Supernovae for an Accelerating Universe and a Cosmological Constant.” *Astron. J.* **116**, 1009 (1998).
3. Betoule, M., et al. “Improved cosmological constraints from a joint analysis of the SDSS-II and SNLS supernova samples.” *Astron. Astrophys.* **568**, A22 (2014).

BAO

4. Beutler, F., et al. “The 6dF Galaxy Survey: Baryon Acoustic Oscillations and the Local Hubble Constant.” *Mon. Not. R. Astron. Soc.* **416**, 3017 (2011).
5. Ross, A. J., et al. “The Clustering of the SDSS DR7 Main Galaxy Sample – A 4% Distance Measure at $z = 0.15$.” *Mon. Not. R. Astron. Soc.* **449**, 835 (2015).
6. Alam, S., et al. (BOSS Collaboration). “The Clustering of Galaxies in the Completed SDSS-III BOSS Survey: Cosmological Analysis of the DR12 Galaxy Sample.” *Mon. Not. R. Astron. Soc.* **470**, 2617 (2017).
7. Eisenstein, D. J., & Hu, W. “Baryonic Features in the Matter Transfer Function.” *Astrophys. J.* **496**, 605 (1998).
8. Eisenstein, D. J., et al. “Detection of the Baryon Acoustic Peak in the Large-Scale Correlation Function of SDSS Luminous Red Galaxies.” *Astrophys. J.* **633**, 560 (2005).
9. Percival, W. J., et al. “Baryon Acoustic Oscillations in the Sloan Digital Sky Survey Data Release 7 Galaxy Sample.” *Mon. Not. R. Astron. Soc.* **401**, 2148 (2010).

CMB / Cosmological Parameters

10. Planck Collaboration. “Planck 2018 Results. VI. Cosmological Parameters.” *Astron. Astrophys.* **641**, A6 (2020).

Distance Measures / Cosmology Foundations

11. Hogg, D. W. “Distance Measures in Cosmology.” arXiv:astro-ph/9905116 (1999).
12. Weinberg, D. H., et al. “Observational Probes of Cosmic Acceleration.” *Phys. Rep.* **530**, 87 (2013).

Optional (if you want to include theoretical background citations)

13. Peebles, P. J. E. “Principles of Physical Cosmology.” Princeton University Press (1993).
14. Mukhanov, V. “Physical Foundations of Cosmology.” Cambridge University Press (2005).